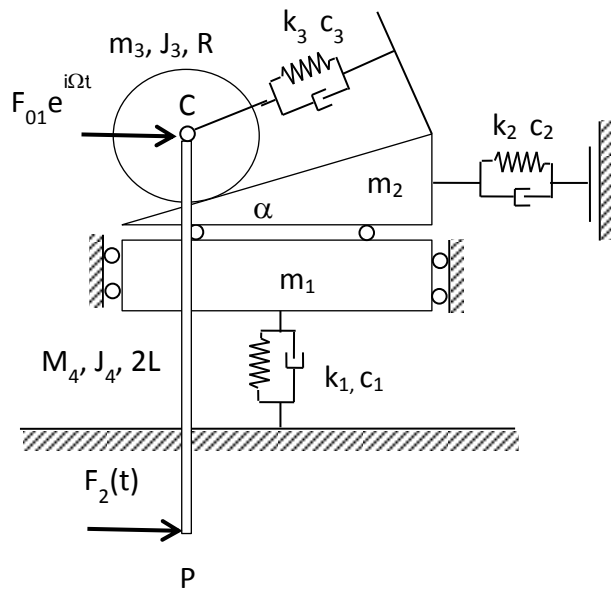


Mechanical Systems Dynamic

February 05th, 2015



$m_1 = 10 \text{ kg}$	$k_1 = 3000 \text{ N/m}$	$L = 0.50 \text{ m}$
$m_2 = 5 \text{ kg}$	$k_2 = 1000 \text{ N/m}$	$R = 0.20 \text{ m}$
$m_3 = 3 \text{ kg}$	$k_3 = 1000 \text{ N/m}$	$\alpha = 30^\circ$
$m_4 = 3 \text{ kg}$	$c_1 = 2 \text{ Ns/m}$	$T = 0.35 \text{ s}$
$J_3 = 0.05 \text{ kgm}^2$	$c_2 = 0.5 \text{ Ns/m}$	$F_{\max} = 10 \text{ N}$
$J_4 = 0.25 \text{ kgm}^2$	$c_3 = 0.5 \text{ Ns/m}$	

The mechanical system above lays in the vertical plane and is shown in its equilibrium position. It is constituted by two translating masses m_1 and m_2 , by a disk m_3 that rolls without sliding on the mass m_2 , and by a bar m_4 connected with a hinge to the center C of the disk.

The students are requested to:

1. Write the linearized equations of motion around the given static configuration;
2. Compute the natural frequencies and modes of vibrations;
3. Compute the Frequency Response Function (in the range 0 – 5 Hz, step = 0.01 Hz) applying an horizontal force F_{01} at point C. Output: the horizontal displacement of the mass m_2 and the force transmitted by the spring and damper system ($k_3 c_3$). Save the graphical results as SNxxx1.fig and SNxxx2.fig, whit S the first letter of your surname, N the first letter of your name and xxx are the last three digits of your identification number (e.g. for Bruni Stefano 123456: BS4561.fig and BS4562.fig)
4. Compute the time history of the disk rotation and the time history of the horizontal component of the displacement of point P, applying the periodic force F_2 shown below (time step $dt = 0.001 \text{ s}$). Save the graphical results as SNxxx3.fig and SNxxx4.fig;
5. Compute the time history of the constraint forces transmitted between disk m_3 and mass m_2 , applying the periodic force defined at the previous point. Save the graphical results as SNxxx3.fig and SNxxx4.fig;
6. Compute the amplitude $F_{\max}$ of the periodic force that leads to the loss of contact between the disk and the mass m_2 , applying the periodic force defined at the point 4 and assuming that the rolling without sliding condition is still verified. Write the numerical result on the sheets of the solution.

Note: Please collect the matlab script file relative to the numerical solution of the exercise and save it with the name SNxxx.m, where S states for the first letter of the surname, N for the first letter of the name and xxx are the last three digits of the identification number (e.g. for Bruni Stefano 123456: BS456.m). It is mandatory that all of the files (.fig e .m) have to be compressed (please select either the .zip or the .rar format) in a file named SNxxx.zip (o SNxxx.rar), which has to be uploaded to the beep portal of Politecnico di Milano. Each student is requested to:

1. Connect to the site <https://beep.metid.polimi.it>;
2. Enter the portal using its own identification code and password;
3. Select the section *Homework* of the course DYNAMICS OF MECHANICAL SYSTEMS (BRUNI STEFANO);
4. Select the folder *Test Jan. 05 2015*
5. Click on the button *Add* and select *Single document*;
6. Browse the compressed file SNxxx.zip using the button *Browse...* (or *Sfoglia...*)
7. Fill the *Title* field writing its own code SNxxx (e.g. for Bruni Stefano 123456: BS456);

Confirm the procedure clicking on the button *Publish*.

